Wei Cheng Lee

Software Engineer

Hsinchu, Taiwan · weichenglee0313@gmail.com · weichenglee0313

Mai0313

Languages

Chinese (Native)

English (Advanced)

Education

University of California, San Diego, BS in Psychology — Sept 2017 – Sept 2021

Built a foundation in statistical analysis and behavioral data modeling, later transitioning to computer science through rigorous self-directed learning. Cross-disciplinary training in cognition, computation, and systems thinking shaped a career spanning AI research for silicon design and firmware engineering for next-generation SoCs.

Experience

Software Engineer, **Google** – Banqiao, New Taipei, Taiwan — Oct 2025 – present

Driving software-led silicon development for **Google** Pixel SoCs. Focused on pre-silicon validation, firmware stability, and system-level power management (LPM), bridging the gap between hardware bringup and OS deployment.

- Expanded Baremetal Test Framework (TFTF) coverage in pre-silicon environments, decoupling firmware testing from OS readiness and accelerating low-power mode (LPM) development by 4 to 8 weeks.
- Developed and optimized ARM Trusted Firmware (TF-A) and Rust Trusted Firmware (RF-A), introducing mechanisms like deterministic sleep algorithms and exponential backoff to eliminate SMC storms and timer race conditions.
- Investigated and resolved complex system-level stability issues, including AXI bus deadlocks, power-down failures, and APC watchdog timeouts.
- Implemented and validated critical firmware workarounds for hardware errata (e.g., Double CMO for early write acknowledgement flaws) across varying core architectures to prevent memory corruption.
- Facilitated the bringup of next-generation SoC architectures (e.g., P26, CDO) utilizing a dual-track development strategy in TF-A and RF-A to ensure project security and delivery timelines.

Machine Learning Engineer, **MediaTek** – Hsinchu, Taiwan — Aug 2022 – Oct 2025

Spearheaded ML research and industrialization for semiconductor design—thermal simulation, leakage analysis, and multi-agent AI systems.

- Developed and deployed physics-aware deep learning models for SoC thermal simulation, reducing maximum temperature prediction error by 34%.
- Engineered layout-dependent leakage analysis methodology, enhancing pre-silicon estimation accuracy by 10% across multiple product lines.
- Built scalable data analysis and visualization pipelines for IC design, improving decision-making speed and accuracy.
- Led automation of RF/analog circuit design using multi-agent LLM systems, driving Pass@1 success rate to over 84% on complex tasks.
- Published 3 peer-reviewed papers in top conferences (**NeurIPS**, IEEE **VLSI TSA**, IEEE **ICLAD**) as first/co-author.
- Established industry-academia partnerships and secured research sponsorships, supporting talent development and knowledge exchange.
- Mentored junior engineers and interns, fostering open collaboration and a culture of continuous learning.

Publications

MenTeR: A fully-automated Multi-agenT workflow for end-to-end RF/Analog Circuits Netlist Design Jan 2025

Proposed a scalable, fully-automated multi-agent workflow for RF/Analog circuit netlist design, integrating Diagram-Aware Retrieval-Augmented Generation and Chain-of-Stage reasoning. Achieved state-of-the-art results on real-world analog circuit design tasks, validating the potential of AI-assisted EDA systems.
Wei Cheng Lee
arxiv.org/abs/2505.22990 (IEEE International Conference on LLM-Aided Design (**ICLAD**))

A Leakage Analysis Methodology Considering Intra-Cell and Inter-Cell Layout Dependent Effects Jan 2024

A novel methodology for digital logic IP-level leakage analysis, incorporating both intra-cell and inter-cell layout-dependent effects. Improved pre-silicon and post-silicon correlation by over 10% in industrial mixed-Vt designs.
Wei Cheng Lee
ieeexplore.ieee.org/document/10546410 (IEEE **VLSI TSA**)

Fast SoC Thermal Simulation with Physics-aware U-Net Jan 2023

A physics-aware U-Net model for rapid SoC thermal simulation, embedding Fourier-Biot constraints in the loss function. Reduced maximum temperature error by 34% and achieved >100x speedup over commercial CFD tools.
Wei Cheng Lee
neurips.cc/virtual/2023/76089 (**NeurIPS** 2023)

Projects

Pre-Silicon Firmware Validation (TFTF) for Pixel SoCs Oct 2025 – present

Expanding the Baremetal Test Framework (TFTF) for pre-silicon environments across next-generation Pixel SoCs (P26, CDO), enabling firmware verification to run independently of OS readiness.

- Built targeted test coverage for system-level power management (LPM) flows, shortening the critical path by 4 to 8 weeks.
- Designed test harnesses for hardware errata and firmware workaround validation prior to tape-out.
- Partnered with hardware, firmware, and validation teams to align on shared pre-silicon test infrastructure.

ARM Trusted Firmware & Rust Trusted Firmware (RF-A) for Pixel SoCs Oct 2025 – present

Contributing to ARM Trusted Firmware (TF-A) and the Rust Trusted Firmware (RF-A) stack for Pixel SoCs, focused on secure low-power transitions, system-level stability, and hardware errata mitigations.

- Executed a dual-track TF-A/RF-A strategy to balance delivery timelines with long-term firmware security goals.
- Introduced deterministic sleep algorithms and exponential backoff to eliminate SMC storms and timer race conditions.
- Diagnosed and resolved AXI bus deadlocks, power-down failures, and APC watchdog timeouts at the system level.
- Implemented Double CMO workarounds for early write acknowledgement errata across varying core architectures.

Automotive Assistant for Vehicle using multi-agent system Jan 2025 – June 2025

Using multi-agent system to develop an automotive assistant for vehicle control, enabling real-time voice commands and automated task execution. Focused on agent orchestration, executor agents, and chain-of-stage reasoning to enhance user experience and system efficiency.

- Implemented a multi-agent architecture for real-time voice command processing and task execution.
- Developed executor agents for specialized tasks, improving system responsiveness and adaptability.

MenTeR: Multi-Agent Analog Circuit Design System Jan 2025 – May 2025

Led engineering and system integration for MenTeR, an end-to-end automated workflow for RF/analog circuit netlist design based on LLM agents. Focused on cross-agent collaboration, reusable knowledge base construction, and seamless integration with industrial flows.

- Architected and engineered PI, Circuit, and Testbench agents for automation of specification analysis, netlist generation, and simulation validation.
- Built the Circuit Think Tank, a reusable domain knowledge base for design pattern acceleration.
- Significantly reduced manual workload and boosted design success rates on real-world analog circuits.
- Drove practical adoption through industry-academia collaboration.

Task Management System with Multi-Agent Orchestration Jan 2024 – Dec 2024

Multi-Agent system for task management, including task assignment, progress tracking, and resource allocation. Utilizes executor agents for efficient execution of tasks across various domains. Supports delegation of tasks to specialized agents, enhancing system flexibility and adaptability.

- Implemented a robust task management framework with multi-agent orchestration.
- Delegated tasks to specialized executor agents for efficient execution.
- Enabled dynamic task assignment and resource allocation based on agent capabilities.

Leakage Analysis Automation Jan 2024 – June 2024

Developed scalable toolchains and visualization dashboards for automated semiconductor leakage analysis, enabling efficient cross-team data-driven validation and model refinement.

- Automated extraction and modeling of layout-dependent leakage effects, deployed across multiple product lines.
- Enabled real-time silicon measurement analysis for rapid debugging and yield improvement.

AI Thermal Simulation Oct 2022 – June 2023

Core developer for a physics-aware, data-efficient SoC thermal simulation engine. Focused on model architecture, loss function design, data pipeline optimization, and experimental validation.

- Integrated thermal gradient and Laplacian constraints, improving data efficiency and model fidelity.
- Benchmarked system against commercial CFD tools, achieving 34% MTE reduction and 100x speedup.
- Explored and prototyped transformer-based variants for future research directions.

Skills

Firmware & Systems Programming (Advanced): ARM Trusted Firmware (TF-A), Rust Trusted Firmware (RF-A), Baremetal Test Framework (TFTF), Low-Power Mode (LPM), SMC Handling, Deterministic Sleep Algorithms, Exponential Backoff

SoC & Hardware Integration (Advanced): Pre-Silicon Validation, SoC Bringup, Hardware Errata Workarounds, AXI Bus Debugging, APC Watchdog, Power Management, Double CMO

Multi-Agent Systems (Expert): Agent-Oriented Programming, Agent Orchestration, Chain-of-Stage Reasoning, Diagram-Aware Retrieval-Augmented Generation (DA-RAG), Executor Agent, Circuit Think Tank

Physics-Informed Machine Learning (Advanced): Physics-Informed Neural Networks (PINNs), Fourier-Biot Equation, Gradient and Laplacian Modeling, Physics-aware Loss Functions

Programming Languages (Expert): Python, C, C++, Rust, Golang, TypeScript, JavaScript, ARM Assembly, LaTeX

AI/ML Frameworks (Advanced): PyTorch, PyTorch Lightning, TensorFlow

DevOps & Infrastructure (Expert): Docker, BigQuery, Kubernetes, GitHub Actions

Research & Analysis (Advanced): Statistical Modeling, Data Analysis, Empirical Research Methods, Technical Writing

Interests

Firmware & Silicon Systems (Research Focus): Trusted Firmware, Pre-Silicon Validation, Low-Power Systems, SoC Architecture, Hardware/Software Co-Design

Systems Programming (Research Focus): Rust, ARM Architecture, Baremetal Testing, Memory Safety, Real-Time Systems

AI for Semiconductor Design (Research Focus): Physics-Informed Neural Networks, Circuit Design Automation, Thermal Modeling, Leakage Analysis

Multi-Agent & Multimodal AI (Research Focus): Multi-Agent Systems, Multimodal & Reinforcement Learning, Decision Making

Research Methodology (Research Focus): Integrative Tokenization, Knowledge Distillation, Process-Oriented Evaluation, Cross-disciplinary Research

Educational Impact (Community): Knowledge Democratization, Fair Access to AI Tools

Volunteer

Contributor, Open Source Community — Jan 2022 – present

Active contributor to open-source machine learning and semiconductor design tools.

- Contributed to 15+ open-source projects related to AI and EDA tools
- Maintained documentation and tutorials for newcomers

Certificates

Fundamentals of Accelerated Computing with CUDA Python — Dec 2024

References

Jiayi Xu

As Manager of **MediaTek**'s Standard Cell Library Dept. II, I witnessed Wei Cheng Lee architect and deliver a custom terabyte-scale data analysis and visualization platform for IC design — replacing Splunk to overcome its scalability and customization limits, and enabling real-time analysis of SPICE models, sidefile models, and silicon measurement data. His command of scalable data pipelines and performance-critical algorithms, paired with deep engagement with IC design workflows, turned a long-standing tooling gap into a decision-making accelerator for the broader engineering organization.

Yi-Chen Lin

As **MediaTek**'s ML Team Manager, I observed Wei Cheng Lee translate his open-source credibility — built through contributions to AutoGen, LangChain, LangGraph, and MLflow — into tangible team outcomes. He directly engaged AutoGen's co-authors, Chi Wang (DeepMind) and Qingyun Wu (Penn State University), to fast-track fixes aligned with our needs and host technical exchanges at **MediaTek**. He also routinely took on the less visible, high-leverage work of hardening testing and data pipelines, raising productivity across the department.

Yu-Sheng Lin

As project manager for **MediaTek**'s AI Thermal Simulation initiative, I saw Wei Cheng Lee drive the core technical execution behind "Fast SoC Thermal Simulation with Physics-aware U-Net" (**NeurIPS** 2023) — integrating physics-aware thermal gradient constraints into the model architecture, validating data efficiency gains through rigorous experimentation, and delivering production-grade code the team could readily extend. He further broadened our research horizon by prototyping a Swin Transformer V2 variant for thermal image generation, showcasing both depth and strategic initiative.